

***Department of Engineering Management
& Systems Engineering
Doctoral Dissertation Defense***

**Wednesday, August 26th, 2026 at 9:30 a.m. - 11:30 a.m.
Science and Engineering Hall
Room 2600**

PINGFAN HU

B.S. Industrial Engineering, 2015, Pennsylvania State University, State College, PA
M.P.E. Mechanical Engineering, 2018, University of Sydney, Sydney, NSW, Australia

DISSERTATION:

***Sustainable Transportation, Sustainable Research: Consumer Adoption and Grid
Peak-Shaving of Battery Electric Vehicle Smart Charging, Inspiring the surveydown
Open-Source Survey Platform***

FIELD OF STUDY: Systems Engineering

ADMISSION TO DOCTOR OF PHILOSOPHY PROGRAM: Fall 2023

ADVISOR OF THE CANDIDATE'S RESEARCH:

Dr. John Paul Helveston

EXAMINING COMMITTEE:

1. Dr. Caitlin Grady, Department of Engineering Management and Systems Engineering, GWU
2. Dr. Johan René van Dorp, Department of Engineering Management and Systems Engineering, GWU
3. Dr. Eric Hittinger, Department of Public Policy, Rochester Institute of Technology

PRESIDING:

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Abstract of Dissertation

Sustainable Transportation, Sustainable Research: Consumer Adoption and Grid Peak-Shaving of Battery Electric Vehicle Smart Charging, Inspiring the surveydown Open-Source Survey Platform

Battery electric vehicle (BEV) adoption is a central pathway to decarbonizing transportation, but realizing its full climate benefit depends on how charging is coordinated with the electricity grid and on the methodological tools available to study consumer behavior in this transition. This dissertation addresses three connected questions through three studies. Study 1 reports a discrete choice experiment with 1,356 current BEV owners that quantifies how program design attributes shape enrollment in supplier-managed charging (SMC) and vehicle-to-grid (V2G) programs. SMC participants value operational flexibility and modest recurring payments, while V2G participants are more sensitive to monetary incentives, reflecting the more active role of bidirectional power flow. One-time enrollment cash, recurring monthly payments, and operational flexibility, such as guaranteed charging thresholds and override allowances, each contribute differently to enrollment, and we provide an interactive web tool that helps utilities and policymakers combine these levers to maximize participation under realistic budget constraints. Study 2 integrates the empirical enrollment curves from Study 1 into a scenario-based grid simulation across the California Independent System Operator (CAISO) and the New York Independent System Operator (NYISO), sweeping time-of-use load-shifting windows, BEV adoption levels, and the full 0% to 100% range of SMC participation. CAISO achieves up to 5.4 GW of peak reduction, NYISO shows more favorable per-household cost efficiency, and cost efficiency is season-invariant in CAISO but deteriorates in NYISO winters when elevated overnight load compresses the off-peak valley window. Diminishing returns to enrollment are sharp in both regions, and enrollment beyond a threshold reconstitutes a new overnight peak on approximately 15.0% of CAISO days, showing that moderate enrollment of 30% to 50%, rather than maximum participation, is the cost-effective operating target. Study 3 introduces surveydown, an open-source markdown-based R package that uses Quarto and Shiny to enable programmable, reproducible surveys with full version control and disaggregated data storage, addressing reproducibility and licensing limitations of existing platforms and making sophisticated experimental designs broadly accessible. Together, these studies translate consumer preferences into utility program design guidance, quantify the regional and seasonal limits of grid-scale peak shaving, and contribute open-source methodological infrastructure to the broader survey research community, advancing both the practical implementation of sustainable transportation and the rigor of the research that supports it.